# Secure Cloud Storage Management using AWS S3 Buckets

## 1. Aim

To create, secure, and delete Amazon S3 buckets using AWS CLI, demonstrating secure cloud storage management.

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## 2. Objective

- To understand cloud storage using Amazon S3
- To create and manage S3 buckets using command-line interface
- To apply security features like blocking public access, versioning, and encryption
- To delete S3 buckets and manage their lifecycle

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## 3. Tools & Technologies Used

- Amazon Web Services (AWS)
- AWS CLI
- Amazon S3 (Simple Storage Service)
- Windows Command Prompt

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## 4. Prerequisites

- AWS Free Tier Account
- IAM User with S3 access
- AWS CLI installed and configured

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## 5. Procedure

**Step 1: Configure AWS CLI**

Configured AWS CLI using IAM credentials.

```
aws configure
```

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**Step 2: Create S3 Buckets**

Created two S3 buckets.

```
aws s3 mb s3://my-first-s3-bucket-1806
aws s3 mb s3://my-second-s3-bucket-1806
```

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**Step 3: Verify Buckets**

Listed all buckets to confirm creation.

```
aws s3 ls
```

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**Step 4: Block Public Access**

Blocked all public access to ensure security.

```
aws s3api put-public-access-block --bucket my-first-s3-bucket-1806 --public-access-block-configuration
BlockPublicAcls=true,IgnorePublicAcls=true,BlockPublicPolicy=true,RestrictPublicBuckets=true
```

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**Step 5: Enable Versioning**

Enabled versioning for data protection.

```
aws s3api put-bucket-versioning --bucket my-first-s3-bucket-1806 --versioning-configuration
Status=Enabled
```

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**Step 6: Delete Second Bucket**

Deleted the second bucket.

```
aws s3 rb s3://my-second-s3-bucket-1806
```

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**Step 7: Suspend Versioning**

Suspended versioning before deletion.

aws s3api put-bucket-versioning --bucket my-first-s3-bucket-1806 --versioning-configuration Status=Suspended

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**Step 8: Delete First Bucket**

Deleted the first bucket.

aws s3 rb s3://my-first-s3-bucket-1806

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**Step 9: Verify Deletion**

Confirmed all buckets were deleted.

aws s3 ls

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## 6. Output

- Two S3 buckets were successfully created
- Security features (public access block, versioning, encryption) were applied
- Buckets were successfully deleted
- No buckets remained after final verification

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## 7. Result

The project was successfully implemented. Secure cloud storage management was demonstrated using AWS S3 buckets and AWS CLI.

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## 8. Conclusion

This project demonstrates how to securely manage cloud storage using Amazon S3. By applying security features such as blocking public access, enabling versioning, and encryption, data is

protected from unauthorized access and accidental loss. AWS CLI provides an efficient way to manage cloud resources through commands.

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## 9. Advantages

- Secure data storage
- Easy management using CLI
- Data protection with versioning
- Encryption ensures confidentiality
- Scalable cloud storage

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## 10. Applications

- Cloud backup systems
- Secure file storage
- Data archiving
- Web application storage
- Enterprise cloud solutions